Nonce Nursery

Cryptography



I started by checking `data.json`, which contained 11 ECDSA signatures (r_i, s_i) generated over the secp256k1 curve and an AES-GCM sealed ciphertext.

Analyzing `chall.py` revealed that nonces were generated using an affine recurrence $k_{i+1} = a \times k_i + b \bmod n$ across a sliding window of 5 consecutive transactions.

Expressing each nonce as $k_i = \alpha_i + \beta_i d \bmod n$, I set up a system of second-order difference equations $\Delta_{j+1} \Delta_{j-1} - \Delta_j^2 \equiv 0 \bmod n$. Eliminating quadratic terms yielded a linear relation that directly solved for private key d.

Finally, I tested the recovered flag against the executable / service:

`python3 Nonce_Nursery.py`

Affine nonce sequence identified at window 0..4.

Flag Captured

`SecLeaf{4ff1n3_n0nc3s_b3nd_th3_curv3}`

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